

The Weberian Economy: How Institutional Persistence Shapes Local Entrepreneurship

Abstract

Counties where religious institutions persisted across generations generate roughly 5.7% more new establishment births per capita even after conditioning on historical income, education, and population scale. Using a new church persistence index that combines standardized church densities weighted by cross-year agreement to emphasize enduring institutional presence, I show that U.S. counties where religious organizations persisted across generations exhibit significantly higher rates of new establishment formation in 2022. The results are robust to rich historical controls, state fixed effects, and alternative index constructions. Mechanism tests suggest that persistence fostered human capital: it is positively correlated with modern educational attainment, and part of its effect on entrepreneurship is mediated by education. Critically, 1890 church density alone does not predict modern entrepreneurship once long-run continuity is controlled, challenging the deep-roots literature's emphasis on initial conditions as the key developmental determinant, a result that also challenges the deep-roots literature's emphasis on initial conditions. At the same time, institutionally persistent counties tended to experience less population growth; they developed greater entrepreneurial capacity per resident, evidence of development through depth rather than scale. Together, the evidence points to a durable pathway through which embedded local institutions accumulated skills, trust, and networks that still shape the entrepreneurial capacity of communities today.

1 Introduction

Consider two U.S. counties with identical populations, identical 1989 median household incomes, and identical 1990 educational attainment. The only systematic difference between them is whether their religious institutions were continuously present across the twentieth century. In 2022, the county with higher church persistence generates approximately 5.7% more new business establishments per capita. This paper asks why and finds that the answer lies not in current religiosity, not in how religious a place was at a single historical moment, but in whether its civic institutions endured continuously across generations, quietly accumulating the human capital and social infrastructure that underpin entrepreneurial activity.

Agglomeration models predict that firm formation concentrates where talent, capital, and market density converge (Glaeser et al. 1992; Duranton and Puga 2004). On this view, rural counties with stagnant populations should be entrepreneurially thin, they lack the density, the labor market thickness, and the knowledge spillovers that cities generate. Yet many such counties consistently outperform their demographic weight in new establishment births. Scale-based theories have no ready explanation for this. The present paper proposes one: institutional depth, not demographic breadth, can sustain local entrepreneurial capacity across generations, particularly in rural economies where place-bound civic organizations historically substituted for the infrastructure that cities take for granted.

Following a Weberian view of economic development, this paper examines how persistent religious institutions, churches that remained active and embedded across generations cultivated the human capital and civic norms that underpin modern entrepreneurship. The mechanism is not doctrinal. As Becker and Woessmann (2009) demonstrate in 19th-century Germany, the economic advantage associated with religious institutions operated primarily through literacy and schooling rather than theology per se. In rural American counties, churches occupied a similar role: they financed local schools, ran literacy programs, normalized educational aspiration, and provided reliable platforms for coordination and trust. Where such institutions endured, these investments compounded across cohorts. The result is not a larger population but a denser entrepreneurial

capability per resident.

To test this argument, I construct a novel county-level church persistence index linking historical congregation counts from the Association of Religion Data Archives (ARDA) across 1890, 1990, 2010, and 2020, weighted by their cross-decade agreement to emphasize genuine institutional continuity over noise. The primary outcome is new establishment births per capita from the Census Business Dynamics Statistics (BDS) in 2022, a measure that captures entrepreneurial intensity at the local level, avoids the compositional problem of net entry, and is consistently defined across geographies. Unlike GDP or employment growth, establishment births reflect the conversion of local capabilities into new firms, which is precisely what a human capital and civic infrastructure story predicts.

The results reveal a consistent pattern across specifications. Counties with higher church persistence exhibit significantly more new establishment births per capita in 2022, but not larger populations or systematically higher incomes. The persistence effect operates primarily through a human capital channel: persistence strongly predicts modern educational attainment, historical schooling predicts firm births, and the persistence coefficient attenuates when contemporary education is added to the model. Crucially, 1890 church density alone does not predict entrepreneurship once long-run continuity is controlled. This finding distinguishes institutional persistence from initial conditions and challenges the conventional deep-roots emphasis on historical starting points as the key developmental determinant.

Results are robust to alternative index constructions (including a two-period 1890–1990 version), inclusion of contemporaneous church density, drop-one-control exercises, substitution of historical for modern population controls, and removal of state fixed effects. Across all specifications, persistence predicts entrepreneurial intensity per capita while exhibiting no positive association with population growth.

This paper makes three contributions to the literature on institutions and entrepreneurship. First, it introduces a multi-period church persistence index that isolates long-run institutional continuity from both initial religious intensity and contemporaneous religious activity a distinction

absent from prior U.S. empirical work. Second, it demonstrates that continuity, not initial conditions, is the operative mechanism: 1890 church density alone predicts nothing, but the persistence index predicts significantly more new establishment births, even when both enter the same regression. This result stands in contrast to much of the deep-roots literature, which treats initial historical conditions as the primary determinant of long-run development (Becker and Woessmann 2009; Nunn 2009), and suggests instead that what institutions do over time matters more than how intense they were at the start. Third, it identifies education as a partial mediator of the persistence–entrepreneurship relationship while leaving a residual direct effect that likely reflects trust, social networks, and coordination capacity providing a richer account of how embedded civic institutions shape local entrepreneurial ecosystems than GDP-focused or religiosity-based measures typically reveal.

By identifying this pathway from institutional persistence, through education, to entrepreneurship, this paper contributes to the growing literature on the deep roots of economic development. The paper proceeds as follows. Section II reviews related work on institutions, human capital, and local economic development, with particular attention to the gap this paper fills. Section III describes the data and the construction of the persistence index. Section IV presents the main empirical findings. Section V reports robustness checks. Section VI discusses the human capital and civic infrastructure mechanisms. Section VII addresses counterarguments and causal limitations. Section VIII concludes with implications for rural development policy and directions for future identification strategies.

2 Literature Review

Three streams of research bear directly on this paper’s argument, and each leaves a specific gap that motivates the present study. The first stream links religious traditions to economic behavior through doctrine and culture (Weber 1905; Becker and Woessmann 2009), emphasizing initial historical conditions. The second links contemporaneous church density to small business activity

and firm survival in U.S. counties (Deller et al. 2018; Conroy and Deller 2020), focusing on current levels. The third and most proximate finds that higher religiosity in U.S. counties is associated with lower GDP growth, interpreted as churches stabilizing incumbents at the expense of dynamic entry (Petach and Powell 2023). What none of these accounts addresses is the temporal dimension of institutional presence: not whether a place is religious now, or was religious at a single historical moment, but whether its religious institutions endured continuously across generations. This gap between initial conditions and contemporaneous levels, between religious intensity and civic continuity is the entry point for the present paper.

A long-standing question in economics is why some places prosper while others do not, even when they share similar resources, legal systems, and market access. Classical development theory emphasized geography, endowments, and trade access, while more recent studies have turned toward institutions, norms, and the long-run imprints of history. The contemporary view, increasingly supported by empirical work, is that much of what determines economic performance today lies not only in present-day policies or investments, but in durable cultural and institutional structures that shape behavior and opportunity across generations.

A related strand of research emphasizes that economic activity is highly persistent across space, even in the presence of large shocks. Studies of path dependence show that historical advantages or institutions can anchor long-run development trajectories. Bleakley and Lin (2012) document that portage sites used in early U.S. transportation networks continue to predict modern urbanization patterns. Redding and Sturm (2008) show that the division and reunification of Germany generated long-lasting effects on city growth, while Davis and Weinstein (2002) find remarkable persistence in the spatial distribution of Japanese cities despite extensive wartime destruction. Together, this literature suggests that local economic equilibria are shaped by slow-moving, place-based forces, providing a natural foundation for studying institutional persistence at the county level.

Human capital plays a central role in many of these persistence mechanisms. Models of endogenous growth emphasize human capital externalities as a key driver of local productivity and development (Lucas, 1988). Empirically, Glaeser et al. (1995, 2004) and Moretti (2004) show

that concentrations of educated workers generate spillovers that raise wages, innovation, and firm formation in surrounding areas. A parallel literature treats entrepreneurship as an inherently local phenomenon, shaped by social networks, skills, and institutions embedded in place. Glaeser, Kerr, and Kerr (2015) and Delgado, Porter, and Stern (2010) emphasize that entrepreneurial activity reflects local capabilities and historical specialization, while evidence from rural contexts suggests that these dynamics can persist even in the absence of large urban agglomerations. This work motivates the use of establishment births as a measure of localized entrepreneurial intensity.

Once established, institutions can deeply embed themselves in local cultures, governance systems, and economic relationships, often proving surprisingly durable. Acemoglu et al. (2001, 2005) argue that the quality of early colonial institutions shaped development paths by establishing incentives that either promoted or slowed productivity and innovation. Nunn (2009) pushes this logic further, showing that even institutions that no longer formally exist, such as those tied to the transatlantic slave trade can leave enduring marks on trust, cooperation, and economic outcomes. These arguments suggest that institutions, whether formal or informal, can cast long shadows, their norms and organizational patterns influencing economic life long after the original context has faded.

Religious institutions are a particularly clear example of this persistence. Max Weber's (1905) *The Protestant Ethic and the Spirit of Capitalism* was one of the first attempts to link religion and economic behavior in a systematic way. He argued that certain Protestant traditions fostered habits of discipline, delayed gratification, and literacy that laid a cultural foundation for economic growth. Though Weber's account was more sociological than empirical, it established a foundational idea: religious traditions can foster not just belief, but embedded norms of work, trust, and education that provide a framework for economic activity. In many communities, especially rural ones, churches have acted as parallel or substitute sources of social coordination, providing stability and investing in human capital when other institutions were absent or weak. Their spatial rootedness, long-run continuity, and civic role make them ideal for studying how institutional continuity might shape long-term development.

Empirical work from other contexts supports this view. Robert Putnam's (1993) study of civic traditions in Italy found that regions with higher historical densities of associations, including religious congregations, displayed stronger democratic performance and better public service delivery in the modern era. His central insight, that institutional density in the past predicts civic and economic capacity in the present, mirrors the ideas explored here, though in a different national setting. Guiso et al. (2006) find that religious affiliation is positively correlated with trust, work ethic, and thrift across European countries, suggesting that religion can shape economically relevant preferences and behaviors. Becker and Woessmann (2009) revisit Weber's thesis through a quantitative lens, arguing that the economic advantage associated with Protestantism in 19th-century Germany came not from doctrine itself, but from its promotion of literacy and schooling, a clear human capital channel.

In the US, scholars have examined the role of religion as a form of local social capital. Deller et al.(2018) document a positive relationship between religious density and small business activity, particularly in non-urban counties. Conroy and Deller (2020) find that higher levels of religious social capital are associated with stronger small business survival rates, interpreting churches as institutional anchors that facilitate coordination, reduce transaction costs, and provide informal support networks for entrepreneurs.

Yet this stabilizing role may come with trade-offs. Petach and Powell (2023), in one of the most closely related studies, find that U.S. counties with higher contemporary church density or religious adherence from 2000 to 2020 experienced lower GDP growth over the same period. They interpret this pattern as evidence that religious social capital may bolster firm survival while dampening productivity growth by reducing entry and exit, thereby lowering regional dynamism. Their findings highlight an important distinction between institutional stability and economic fluidity, they raise a specific challenge for any religion and entrepreneurship paper. The present study addresses this challenge directly. The Petach and Powell finding and the persistence result here are not contradictory; they reflect different temporal objects. Short-run religiosity may indeed slow creative destruction by stabilizing incumbents. Long-run institutional continuity may operate

through a different channel entirely, building the human capital and social infrastructure that make entrepreneurial entry possible in the first place. The two findings can coexist if the economic effects of religious institutions depend critically on the time horizon over which they are measured, a distinction the present paper formalizes through the persistence index.

This perspective builds on established arguments that religious institutions matter economically through human capital formation and the accumulation of civic norms, rather than doctrine itself (Becker and Woessmann, 2009; Putnam, 1993). By emphasizing institutional continuity rather than contemporaneous religious activity, this approach situates religious life within the broader literature on the deep roots of development. In many rural American counties, churches historically functioned as central civic anchors, shaping education, norms, and local networks over generations. A persistence framework therefore complements existing studies by suggesting how slow-moving, place-based institutions may influence modern economic outcomes through both human capital accumulation and non-educational channels such as trust, coordination, and social networks.

A large empirical literature measures local entrepreneurship using new establishment births from the Census employer business demography (BDS/CBDS). Haltiwanger et al. (2013) formalize the role of young businesses in job creation and operationalize entry with establishment births in the BDS. Decker et al. (2014) use the same births-based entry to document the contribution of new firms to U.S. job creation and to the secular decline in dynamism. Fort et al. (2013) employ BDS entry to study how firm age mediates cyclical responses. Pugsley and Şahin (2019) analyze startup rates over the business cycle with BDS/CBDS entry series, arguing that fluctuations in entry are central to macro dynamics. Outside strictly academic outlets, Hathaway and Litan (2014) popularize births-based entry rates to compare dynamism across states and metros. These studies justify establishment births as a standard, policy relevant indicator of local entrepreneurial activity: it is consistently defined across places, captures the intensive margin of new firm formation, and aligns with theories that link institutions and human capital to the creation (not just survival) of new businesses.

This paper extends that conversation by focusing on institutional persistence over more than

a century, asking not whether a county is currently religious, or was densely churched in 1890, but whether its religious infrastructure endured continuously across generations. The persistence index operationalizes this distinction. By measuring establishment births per capita rather than GDP, the paper is sensitive to the intensive-margin entrepreneurial activity that may flourish in smaller, slower-growing counties even as it goes undetected in aggregate income measures. Together, these choices allow the paper to test a claim that the existing literature has not directly examined: that institutional continuity, independent of initial conditions and contemporary levels, shapes the entrepreneurial capacity of communities today.

3 Data and Empirical Strategy

3.1 Data Sources

This study combines historical and contemporary county-level records to trace how the long arc of religious institutions relates to modern economic life. The primary source is the Association of Religion Data Archives (ARDA), which reports county congregation counts from 1890 through 2023. This data spans denominations and provides a consistent, spatially granular measure of religious infrastructure across time. Economic and demographic measures are drawn primarily from the U.S. Census and related federal series via NHGIS. Entrepreneurship is measured using establishment births from the Business Dynamics Statistics (BDS), defined as when a firm reports positive employment for the first time. The data and exact years used are as follows:

- Churches per 1,000 people in 1890, 1990, 2010, and 2020
- median household income in 1989 and 2010 (adjusted for inflation to constant 2023 dollars);
- Educational attainment in 1990, 2010, and 2023, (share of adults aged 25+ with at least a high school diploma);
- Population in 1890, 1990, 2010, and 2023;

- New establishment births in 1989, 2010, and 2022.

The years chosen reflect the availability of church counts and other relevant data. Although intermediate benchmarks (1906–1980) exist, they suffer from severe coverage gaps and comparability problems. These issues, together with declining participation, led the Census Bureau to discontinue its religious bodies survey after 1936. I include 1890 despite its noisiness because it is the only near-comprehensive national enumeration of local religious presence before the federal Census of Religious Bodies began in 1906. It anchors the index in a historical period when churches functioned as central civic institutions at the county level. Robustness checks indicate that the index’s predictive content is driven primarily by variation in the modern decades (1990–2020) rather than by the 1890 measure alone, with the early data serving mainly to establish a long-run institutional baseline rather than the main driver of results. The 1990 ASARB census marks the first comprehensive, standardized national enumeration since the 1936 cutoff, and subsequent rounds in 2010 and 2020 provide consistent modern benchmarks. Baseline controls line up with this timing: household income is consistently measured at the county level for 1989, educational attainment in 1990 (1990 Census), and establishment births in the earliest county-consistent BDS collections roughly 1989–1990. Final outcomes use 2022–2023 data (BDS, ACS/Census).

County coverage is nearly complete, with a few notable exceptions. Connecticut and Rhode Island lack consistent, functional county governments and are therefore excluded. Alaska’s borough and census area system does not fully align with historical county definitions, limiting comparability in early years. Historical coverage is also incomplete for a small subset of counties for example most in Oklahoma and Montana in 1890 due to gaps in early church counts and population records. In addition, a small number of sparsely populated rural counties are missing from the dataset due to incomplete reporting. As a result, final estimation samples range from roughly 2,600 (82%) to just over 3,000 (95%) counties, covering the vast majority of the continental United States. (Appendix D)

3.2 Key Variables

Church persistence To capture long-run institutional continuity, I construct a four-period church persistence index drawing on county-level church densities from 1890, 1990, 2010, and 2020 (ARDA). The index weights each decade by its average agreement with the other decades, so years with more consistent cross-county rankings contribute more to the composite. A county with a high score maintained above-average church density not in one period but across all four it is the consistency of institutional presence over 130 years, not a single peak of religious activity, that the index is designed to capture.

Let $t \in \mathcal{T} = \{1890, 1990, 2010, 2020\}$ and let $CD_{i,t}$ be church density (per 1,000) for county i in year t . For each county i :

- (i) *Each year of church counts is standardized.* Let $m_t = \text{median}_i(CD_{i,t})$ and $MAD_t = \text{median}_i(|CD_{i,t} - m_t|)$. Define the robust z for year t :

$$\tilde{Z}_{i,t} = \frac{CD_{i,t} - m_t}{MAD_t}.$$

- (ii) Extreme values are winsorized at ± 3.5 .

- (iii) *Spearman-weighted composite.* Let $\rho_{t,t'}$ be the cross-county Spearman correlation between $\tilde{Z}_{\cdot,t}$ and $\tilde{Z}_{\cdot,t'}$. Set the weight for year t to its average agreement with other years,

$$\omega_t = \frac{1}{|\mathcal{T}| - 1} \sum_{t' \in \mathcal{T} \setminus \{t\}} |\rho_{t,t'}|, \quad w_t = \frac{\omega_t}{\sum_{s \in \mathcal{T}} \omega_s}.$$

Let $\mathcal{T}_i \subseteq \mathcal{T}$ denote the set of years observed for county i . When a county is missing observations in some years, weights are re-normalized over the available years, and the composite is formed as

$$S_i = \sum_{t \in \mathcal{T}_i} \frac{w_t}{\sum_{s \in \mathcal{T}_i} w_s} \tilde{Z}_{i,t}.$$

(iv) *Re-standardization across counties.*

$$\text{Persistence}_i = \frac{S_i - \bar{S}}{\text{sd}(S)}.$$

so that the composite is standardized across counties (mean 0, s.d. ≈ 1). A high score indicates a county with consistently above-average church density across multiple decades.

Outcomes Four variables measure contemporary economic outcomes:

1. **Entrepreneurship (primary):** $\log(1 + \text{NewEstab}_{2022})$. This is a count transformed with $\log(1 + y)$ to retain zero-birth counties and stabilize skew
2. **Education:** High-school-or-more attainment share in 2023 (percentage points)
3. **Population:** $\log(\text{Population}_{2023})$

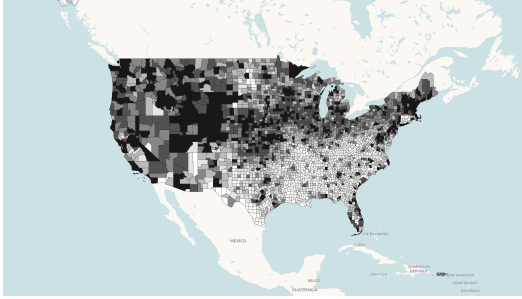
Controls To separate long-run persistence from baseline conditions, the preferred specifications include:

- Historical economic level: Median household income (1989; standardized);
- Historical human capital: educational attainment (1990; standardized);
- Population scale: $\log(\text{Population}_{2010})$;
- State fixed effects (δ_s) to absorb persistent state-level factors.

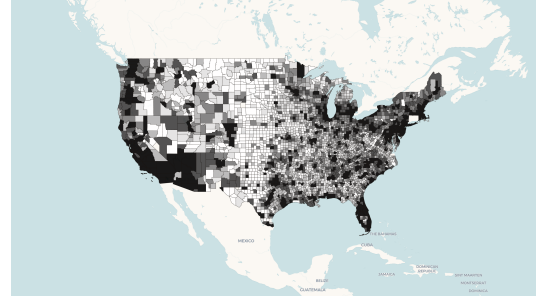
Heatmaps The figure below visualizes the spatial distribution of the persistence index and key outcomes. Panel (a) shows that institutional persistence is concentrated in the South and parts of the Midwest, reflecting regions where church density remained comparatively stable over the long run. Panel (b) displays the expected clustering of population in large metropolitan and coastal counties, a pattern that differs from the persistence distribution. Panel (c) indicates that new establishment births per capita are relatively strong in smaller, non-urban counties rather than in the

most populous regions. Panel (d) shows that counties with higher persistence also tended to exhibit higher educational attainment by 1990.

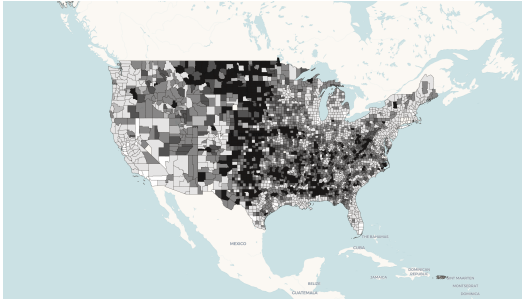
Taken together, these maps suggest that persistence is not simply capturing urbanization or population scale. Instead, persistence aligns more closely with historical human capital formation and with contemporary entrepreneurial intensity in less urbanized areas, consistent with a mechanism operating through localized, long-run institutional and educational development rather than through sheer population growth.



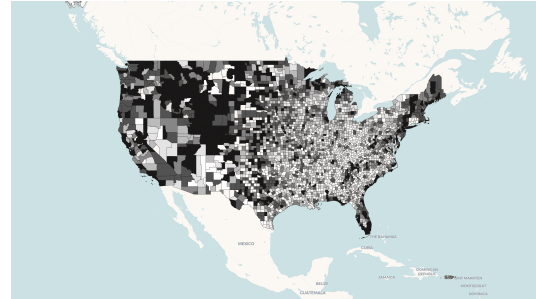
(a) Persistence Heatmap



(b) Population 2023 Heatmap



(c) New Establishment density 2022 Heatmap



(d) Education 1990 Heatmap

(Darker shading indicates higher values) County level heatmaps suggest a transmission channel: persistence aligns more closely with historical education than directly with new establishments, consistent with the mediation results. Counties with high persistence often show stronger educational attainment, which in turn education is associated with higher contemporary entrepreneurial intensity per capita.

- **Persistence index:** $[-1.667, -0.9797], [-0.9797, -0.5015], [-0.5015, -0.2925], [-0.2925, 0.3939], [0.3939, 0.8433], [0.8433, 12.28]$.
- **Population (2023):** $[81, 7,581], [7,581, 15,150], [15,150, 26,140], [26,140, 46,830], [46,830, 121,600], [121,600, 9.663\text{M}]$.
- **Birth density (2022):** $[0, 1.328], [1.328, 1.622], [1.622, 1.857], [1.857, 2.152], [2.152, 2.686], [2.686, 11.38]$.
- **Education (1990, % HS+):** $[18.1, 57.7], [57.7, 64.8], [64.8, 71.3], [71.3, 75.2], [75.2, 79.4], [79.4, 95.5]$.

Table 1 Summary Statistics

Variable	Mean	Std. Dev.	Min	Max	N
Persistence Index	0.00	1.00	-1.72	3.44	3,055
Persistence Index (1890–1990)	0.00	1.00	-2.23	2.48	3,055
New Establishments, 2022	273.5	1,113.2	0.00	38,467.0	3,004
Population, 1890	22,257.55	54,068.18	0	1,515,301	2,716
Population, 2010	98,232.7	315,733.8	90.0	9,818,605.0	3,055
Population, 2023	106,664.0	335,829.0	81.0	9,663,345.0	3,055
Median Household Income, 1989 (real, \$2023)	58,136.0	15,572.0	21,016.0	144,959.0	3,055
Educational Attainment, 1990 (%)	68.80	11.90	18.10	95.50	3,055
New Establishments, 1989	231.4	837.9	0.00	26,540.0	3,013

3.3 Regression Specifications

The empirical strategy proceeds in three steps. First, I estimate a baseline association between long-run religious persistence and modern entrepreneurship, conditioning on historical economic and human capital levels, population scale, and state context. Second, I examine alternative outcomes to probe mechanism and rule out simple scale stories. Third, I evaluate robustness to alternative institutional measures and control sets.

For county i in state s , I estimate the following baseline specification:

$$Y_{is} = \beta_0 + \beta_1 \text{Persistence}_i + Z_i' \gamma + \delta_s + \varepsilon_{is},$$

where δ_s denotes state fixed effects. All models are estimated with HC3 standard errors, and historical control variables are standardized prior to estimation.

The dependent variable Y_{is} alternately measures log new establishment births in 2022, educational attainment in 2023 (the share of adults aged 25+ with at least a high school diploma), log population in 2023.

The set of control variables varies by outcome. When Y_{is} is log new establishment births in 2022 and educational attainment in 2023, the control vector is

$$Z_i = (\text{Income}_{i,1989}^{\text{std}}, \text{Education}_{i,1990}^{\text{std}}, \log(\text{Pop}_{i,2010})),$$

which adjusts for historical economic conditions, baseline human capital, and county scale. When Y_{is} is log population in 2023, the control vector excludes population and is given by

$$Z_i = (\text{Income}_{i,1989}^{\text{std}}, \text{Education}_{i,1990}^{\text{std}}),$$

so that population is not mechanically controlled for when it is the outcome of interest.

Why these models?

- Baseline: Tests whether long-run institutional persistence predicts modern entrepreneurship
- Education outcome: Probes the *human-capital channel*: if persistence raises contemporary attainment, that's a necessary step for persistence to matter for firms.
- Population outcome: Rules out simple scale stories. A weak or negative coefficient implies persistence is not just picking up bigger places (and may even be stronger in historically smaller counties).

Education and new establishments directly estimates the link between historical education and modern firm creation, providing evidence that human capital mediates entrepreneurship.

$$\log(1 + \text{NewEstab}_{is,2022}) = \beta_0 + \beta_E \text{Education}_{i,1990}^{\text{std}} + \gamma_1 \text{Income}_{i,1989}^{\text{std}} + \gamma_2 \log(\text{Pop}_{i,2010}) + \delta_s + \varepsilon_{is}.$$

Mediation via education (attenuation test) I compare Eq.1 estimates with and without historical education and then include contemporaneous education directly:

$$\begin{aligned} \log(1 + \text{NewEstab}_{is,2022}) = & \beta_0 + \beta_1 \text{Persistence}_i + \lambda \text{Education}_{i,2023} \\ & + \gamma_1 \text{Income}_{i,1989}^{\text{std}} + \gamma_2 \text{Education}_{i,1990}^{\text{std}} + \gamma_3 \log(\text{Pop}_{i,2010}) + \delta_s + \varepsilon_{is}. \end{aligned}$$

The attenuation of $\hat{\beta}_1$ when Education_{2023} enters is consistent with a human capital channel. I

report the implied proportion mediated in table 4.

Initial conditions vs. continuity To distinguish persistence from simple historical levels, I replace the index with the 1890 church density and also estimate a model including both measures distinguishing long-run institutional continuity from simple historical starting conditions:

$$\begin{aligned} \log(1 + \text{NewEstab}_{is,2022}) = & \beta_0^H + \beta_1^H \text{ChurchDensity}_{i,1890} \\ & + \gamma_1 \text{Income}_{i,1989}^{\text{std}} + \gamma_2 \text{Education}_{i,1990}^{\text{std}} + \gamma_3 \log(\text{Pop}_{i,2010}) + \delta_s + \epsilon_{is}^H. \end{aligned}$$

$$\begin{aligned} \log(1 + \text{NewEstab}_{is,2022}) = & \beta_0^B + \beta_1^B \text{Persistence}_i + \beta_2^B \text{ChurchDensity}_{i,1890} \\ & + \gamma_1 \text{Income}_{i,1989}^{\text{std}} + \gamma_2 \text{Education}_{i,1990}^{\text{std}} + \gamma_3 \log(\text{Pop}_{i,2010}) + \delta_s + \epsilon_{is}^B. \end{aligned}$$

Comparing β_1^H to β_1^B and β_2^B assesses whether long-run continuity (persistence) is more informative than initial intensity alone.

4 Main Results

This section presents the core empirical findings. I begin with the baseline relationship between long-run church persistence and modern new establishment births, then examine the human-capital channel, and finally report results for population as an alternative outcome.

4.1 Core Findings

A. New establishment births (main outcome). Estimating the baseline, the coefficient on the church persistence index is positive and statistically significant: $\hat{\beta}_1 = 0.0559$ ($p = 0.000$; $R^2 = 0.954$; $N = 2,968$).

Table 2 Baseline: Persistence and Establishment Births

	OLS: $\log(1 + \text{NewEstab_2022})$
Persistence Index	0.0559*** (0.014)
Income 1989 (std)	0.0682*** (0.012)
Educational Attainment 1990 (std)	0.2118*** (0.018)
$\log(\text{Population 2010})$	1.0197*** (0.010)
State fixed effects	Yes
Observations	2,968
R^2	0.954

Notes: HC3 robust standard errors in parentheses. Variables labeled “(std)” are standardized to mean 0 and SD 1 prior to estimation. All models include state fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Because the index is standardized a one standard deviation increase in Persistence is associated with a $100 \times (e^{0.0559} - 1) \approx 5.75\%$ increase in new establishment births.

The baseline equation formalizes the specification. The dependent variable is the log of new establishment births in 2022, so coefficients can be interpreted as semi-elasticities. The coefficient of interest is β_1 on the Persistence Index, capturing the long-run association between religious institutional persistence and contemporary entrepreneurship.

Control variables absorb important county characteristics: standardized 1989 income and 1990 education attainment proxy for baseline economic development and human capital; and $\delta_{s(i)}$ are state fixed effects that capture state-level factors such as policy and legal institutions. The key takeaway from the baseline: among counties that are observationally similar today in size, income, and educational history, those whose religious institutions endured across generations consistently generate more new businesses. Scale alone does not explain this.

Under this specification, β_1 measures the conditional difference in new establishment births across counties with differing levels of institutional persistence, holding constant historical economic capacity, human capital, and state environments.

Results are robust to alternative constructions of the persistence index. I re-estimate the index using only the 1890 and 1990 benchmarks to reflect the historical break in coverage and to show

that the results are not driven by the later decades. This version produces similar coefficients across the same models, (baseline with comparison index $\hat{\beta}_1 = 0.0375$ ($p = 0.001$; $R^2 = 0.954$; $N = 2,968$).

B. Education channel Turning to mechanism, using the education outcome under control set A shows that persistence is strongly associated with contemporary human capital: $\text{Education}_{i,2023}$ rises with persistence (coeff = 0.4007, $p = 0.004$; $R^2 = 0.655$; $N = 3,055$; Table 3). Consistent with a transmission from human capital to entrepreneurship, regressing education on contemporary new establishments yields a large and precise association between historical schooling and new establishment births (coeff = 0.2088, $p = 0.000$; $R^2 = 0.954$; $N = 2,968$; Table 3).

Table 3 Education Channel: Persistence, Human Capital, and Entrepreneurship

	(a) Persistence and Education Education_2023	(b) Education and Establishment Births $\log(1 + \text{newEstab}_{2022})$
Persistence Index	0.4007*** (0.141)	
Income 1989 (std)	-0.2184 (0.127)	0.0613*** (0.012)
Educational Attainment 1990 (std)	5.4929*** (0.188)	0.2088*** (0.018)
$\log(\text{Population } 2010)$	-0.2307*** (0.092)	0.9962*** (0.008)
State fixed effects	Yes	Yes
Observations	3,055	2,968
R^2	0.655	0.954

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

To gauge mediation directly, I augment the baseline with contemporaneous education. The persistence coefficient attenuates from 0.0559 ($p = 0.000$) to 0.0509 ($p = 0.000$) when $\text{Education}_{i,2023}$ is included, while $\text{Education}_{i,2023}$ itself is positive and precisely estimated (coeff = 0.0140, $p = 0.000$; Table 4) with implied proportion mediated of 9%. The human capital channel is real but partial: persistence predicts contemporary education, education predicts firm births, and the persistence coefficient attenuates when education enters. That the attenuation is only 9% is itself informative, a substantial share of the persistence effect operates through channels other than for-

mal schooling, likely trust, networks, and civic coordination capacity.

Table 4 Mediation Test: Persistence, Education₂₀₂₃, and New Establishments

	OLS: $\log(1 + \text{NewEstab_2022})$	
	(1) Baseline	(2) + Education ₂₀₂₃
Persistence Index	0.0559*** (0.014)	0.0509*** (0.014)
Income 1989 (std)	0.0682*** (0.012)	0.0717*** (0.012)
Educational Attainment 1990 (std)	0.2118*** (0.018)	0.1342*** (0.021)
Education 2023 (pct HS+)		0.0140*** (0.003)
$\log(\text{Population 2010})$	1.0197*** (0.010)	1.0223*** (0.010)
State fixed effects	Yes	Yes
Observations	2,968	2,968
R^2	0.954	0.955

Notes: Attenuation of the Persistence coefficient when adding Education₂₀₂₃: Persistence coef: 0.0559 $\rightarrow$ 0.0509 (with edu_2023); attenuation = 9.0%. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C. Alternative outcomes regressing persistence on population using control set B indicates a negative association of persistence with $\log(\text{Pop}_{i,2023})$ (coeff = -0.9072 , $p = 0.000$; $R^2 = 0.716$; $N = 2968$). This is not a paradox, it is a signature. Institutionally persistent counties are smaller places that grew less over the twentieth century. The positive persistence entrepreneurship result (controlling for current size) and the negative persistence population result together tell a coherent story: these counties did not grow outward, they grew inward, building more enterprise per person rather than more people.

Table 5 Persistence and Population

	OLS: log(Population_2023)
Persistence Index	-0.9072*** (0.025)
Income 1989 (std)	0.3647*** (0.033)
Educational Attainment 1990 (std)	0.1374*** (0.041)
State fixed effects	Yes
Observations	2,968
R^2	0.716

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Further stress tests, alternative specifications, and diagnostic exercises are presented in the next section and Appendix A.

Initial Conditions vs. Continuity To test whether the results are driven by initial religious intensity rather than continuity, I replace the persistence index with the 1890 church density and also estimate a model including both measures. As shown in Table 6, church density in 1890 alone is not significantly related to modern new establishments ($\hat{\beta}_1 = 0.0003$, $p > 0.10$). When both persistence and density are included, persistence remains significant ($\hat{\beta}_1 = 0.0532$, $p = 0.001$), while the 1890 density is null ($\hat{\beta}_2 = -0.0022$, $p > 0.10$). This result is the most direct challenge to the conventional deep-roots framing. It is not where you started that matters it's whether you kept going. Institutional continuity across generations, not intensity at a single historical moment, is what predicts entrepreneurial capacity today.

Table 6 1890 Church Density, Persistence, and Establishment Births

	(a) 1890 Church Density Only log(1 + NewEstab ₂₀₂₂)	(b) Persistence and 1890 Density log(1 + NewEstab ₂₀₂₂)
Persistence Index		0.0532*** (0.015)
Church Density 1890 (per 1,000)	0.0003 (0.004)	-0.0022 (0.004)
Income 1989 (std)	0.0624*** (0.012)	0.0694*** (0.013)
Educational Attainment 1990 (std)	0.2024*** (0.019)	0.2062*** (0.019)
log(Population 2010)	0.9980*** (0.008)	1.0184*** (0.011)
State fixed effects	Yes	Yes
Observations	2,610	2,610
R^2	0.956	0.956

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Interpretation and Diagnostics Unless noted otherwise, log linear specifications are estimated by OLS with heteroskedasticity robust (HC3) standard errors; statistical significance is evaluated at the 5% level and R^2 is reported. Because coverage varies across historical series, estimation uses complete case (listwise) deletion for the variables entering each model, so the sample size varies by specification.

For log specifications (e.g., new establishments, population), coefficients are interpreted as semi-elasticities: a one-unit increase in X changes Y by $100 \times (e^\beta - 1)\%$ (approximately $100\beta\%$ for small $|\beta|$). For example, with $\beta = 0.0559$ on Persistence, a one-unit increase is associated with a $100(e^{0.0559} - 1) \approx 5.75\%$ increase in new establishments. For models using the church persistence index, coefficients reflect the effect of a one-unit change in the index; when a standardized version is used in robustness checks, coefficients correspond to a one-standard deviation increase. Education outcomes are in percentage point units.

A noteworthy feature of the baseline specification is its high explanatory power ($R^2 = 0.963$). This reflects the strong persistence of county level fundamentals and the inclusion of multiple historical controls (income, education, and population scale), which together explain much of the cross-county variation in establishment density. While such a high R^2 raises concerns about po-

tential overfitting, robustness checks confirm that the persistence coefficient is not mechanically driven by collinearity among the controls.

5 Robustness and Alternative Specifications

This section evaluates the stability of the baseline persistence entrepreneurship relationship by probing alternative specifications, control sets, and institutional measures. Results are summarized below, with full estimates reported in Appendix A.

- (i) **Drop-one-control at a time:** re-estimate the baseline removing each covariate in turn (historical income, historical education, and $\log(\text{Pop}_{2010})$) to gauge the sensitivity of $\hat{\beta}_1$.
- (ii) **Bundled exclusions:** To assess whether the baseline association reflects per-capita intensity versus levels, and to distinguish Persistence from long-run fundamentals, I re-estimate the baseline under two exclusions. First, I omit $\log(\text{Pop}_{2010})$, forcing the model to explain raw establishment counts without dividing out scale. Second, I retain scale but jointly omit the historical fundamentals ($\text{Income}_{1989}^{\text{std}}$ and $\text{Education}_{1990}^{\text{std}}$). This is to show that conditioning on historical fundamentals is necessary to isolate the persistence channel.
- (iii) **Alternative institutional measure:** add $\text{ChurchDensity}_{2020}$ alongside Persistence_i to verify results are not driven by contemporaneous church presence alone.
- (iv) **Replacing $\log(\text{pop } 2010)$ with $\log(\text{pop } 1890)$:** I re-estimate the entrepreneurship specification substituting historical population (1890) for the modern scale control to test whether the persistence coefficient reflects long-run size trajectories rather than contemporary scale. This isolates whether institutional continuity is associated with *initial* county size (and subsequent growth paths) versus *current* scale.
- (v) **Fixed-effects scope:** estimate pooled models without $\delta_{s(i)}$ to show whether the relationship relies on within state comparisons.

- (vi) **Alternative Construction, Two-year index (1890–1990):** Using the same procedure as the four-year persistence index, I construct a simplified two-period version based only on 1890 and 1990 church densities. This serves as a validity check for the main index, testing whether the core results hold when using only the earliest and a modern benchmark year, thus probing robustness to the long temporal gap between historical and contemporary data. Estimates remain positive and statistically significant.

5.1 Control Exclusion Tests

I re-estimate the baseline specification by removing each covariate in turn (Table A1). Removing $\log(\text{Pop}_{2010})$ flips the sign to -0.8241 ($p < 0.001$): without a scale control the model explains raw new establishment birth counts, and because persistence is concentrated in smaller/rural counties, the coefficient loads negatively on size. Dropping Education_{1990} reduces the coefficient to 0.0485 ($p = 0.001$), and omitting Income_{1989} yields 0.0475 ($p = 0.000$), indicating that conditioning on historical human capital and income is important to isolate the persistence entrepreneurship link.

When I remove $\log(\text{Pop}_{2010})$ as shown earlier, the sign to -0.8241 ($p < 0.001$). Conversely, omitting historical income and education while retaining $\log(\text{Pop}_{2010})$ renders persistence insignificant (0.0081 , $p = 0.597$), suggesting the association operates primarily through human capital rather than through contemporaneous scale or the inherited establishment base. Bottom line: the main result depends on conditioning on both population scale and historical fundamentals which is exactly what the human capital mechanism predicts. The sign flip without population control reflects composition, not a substantive reversal.

5.2 Alternative Institutional Measure

Adding contemporaneous church density (2020) alongside the persistence index (Table A3) leaves the persistence effect intact ($\hat{\beta}_1 = 0.1002$, $p = 0.000$), while church density in 2020 is not significant ($\hat{\beta}_2 = -0.0334$, $p = 0.087$). This tells us current levels of religious activity do not explain the persistence result. It is the long arc of institutional presence, not today's church density, that

predicts entrepreneurship.

5.3 Institutional Persistence and the Intensive Margin of Development

The contrast between specifications using historical versus contemporary population controls reveals the distinctive nature of institutional persistence. Table 7 demonstrates that the relationship between church persistence and entrepreneurship fundamentally depends on whether one conditions on initial or final county size.

Table 7 The intensive margin: population controls and entrepreneurship (variant of baseline model)

	(1)	(2)
	Contemporary Control	Historical Control
	log(Pop ₂₀₁₀)	log(Pop ₁₈₉₀)
Persistence Index	0.0559*** (0.014)	-0.6986*** (0.029)
Historical Controls	Yes	Yes
State Fixed Effects	Yes	Yes
Observations	2,968	2,652
R^2	0.954	0.756

Notes: Both specifications include standardized income (1989), education (1990), and state fixed effects. Robust standard errors in parentheses. *** $p < 0.01$.

Column (1) shows that among counties of similar size *today*, those with greater institutional persistence exhibit significantly higher entrepreneurial intensity. Column (2) reveals the complementary fact: counties with persistent religious institutions were historically smaller and experienced less demographic growth over the twentieth century (only when 1890 population is the sole scale control). Essentially, persistence characterizes places that were historically small and stayed small demographically but became disproportionately entrepreneurial given their eventual size. The two coefficients together define 'developmental depth.'

This pattern supports our intensive margin interpretation: institutional persistence characterized places that grew little demographically but learned to do more with the people they had. The negative coefficient with 1890 population indicates these were not naturally entrepreneurial counties in terms of growth trajectories; the positive coefficient with 2010 population demonstrates they became exceptionally entrepreneurial *given their eventual size*.

This evidence aligns with historical accounts of religious institutions as stabilizing forces in rural communities. Rather than fueling population expansion, persistent churches appear to have deepened local human capital and entrepreneurial capacity within existing demographic constraints. The mechanism operated not through scale, but through what we term “developmental depth”, building economic resilience and business formation capability independent of population growth.

As an additional diagnostic, I jointly included both 1890 and 2010 population to account for long run demographic path dependence. The coefficients behave as theory predicts, modern population enters positively, historical population enters negatively, and the persistence index remains positive and significant. This confirms that the main result is not driven by historical settlement patterns or demographic growth trajectories.

5.4 No State Fixed Effects

Dropping state fixed effects (Table-A5) causes the persistence coefficient to shrink substantially to 0.0205 ($p = 0.122$). This suggests that much of the identifying variation comes from within-state comparisons rather than across state differences.

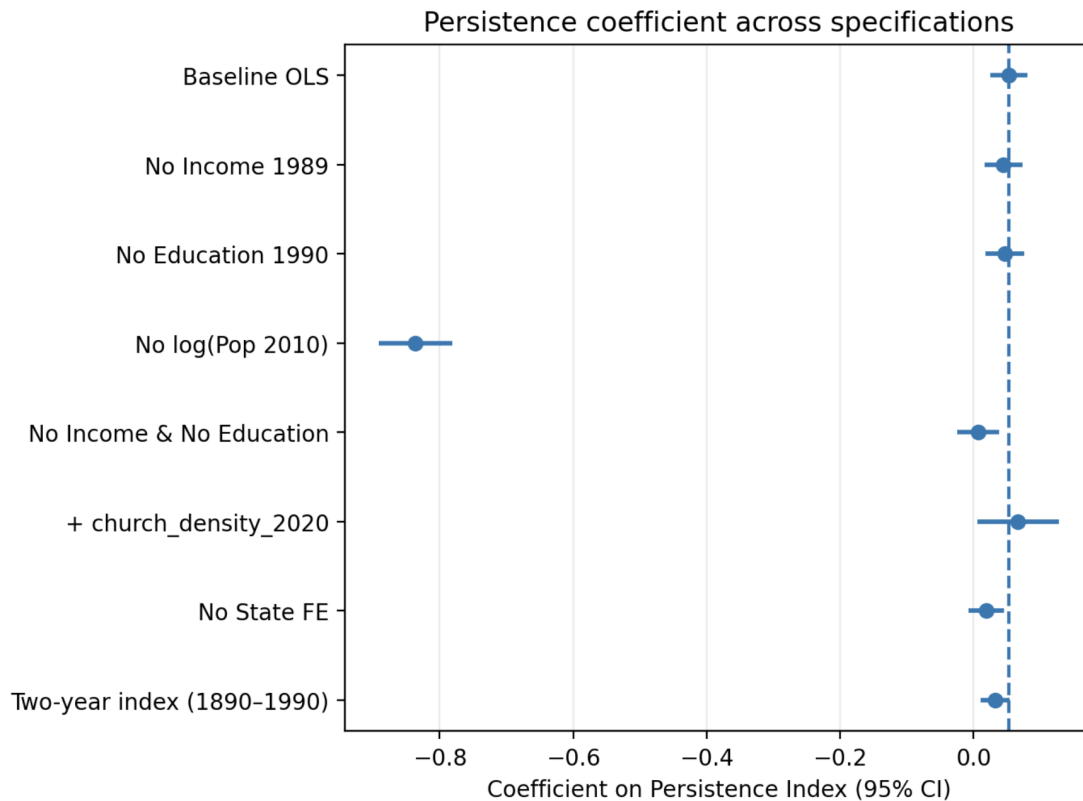
5.5 Residual Diagnostics

Residual checks (Appendix A) indicate that model assumptions are broadly satisfied. Histograms and Q–Q plots show that residuals are approximately normal, with only mild deviations in the tails. Studentized residuals, leverage statistics, and Cook’s distance reveal outliers don’t have an effect on the model; results remain unchanged when re-estimated excluding the highest-leverage observations.

Formal tests (Breusch–Pagan) reject homoskedasticity at conventional levels, but all models are estimated with HC3 heteroskedasticity-robust standard errors, ensuring valid inference. Together, these diagnostics suggest that the results are not driven by outliers or model misspecification.

5.6 Alternative Index Construction (1890-1990)

Estimating Equation 1 with the two-year index produces $\hat{\beta}_1 = [0.0375]$ with a standard error of $[0.011]$ ($p = [0.001]$, $R^2 = [0.954]$, $N = [2,968]$). The magnitude is smaller than the four-year index but remains positive and statistically significant, reinforcing that institutional persistence measured at the turn of the 20th century is linked to modern entrepreneurial activity. (Table A6)



Coefficient Estimates Across Specifications. Points plot the estimated coefficient on the Church Persistence Index and horizontal lines show 95% confidence intervals. The blue dotted line is the main model coefficient.

6 Mechanism Discussion: Human Capital and Embedded Institutions

Three empirical patterns motivate the mechanism discussion. First, persistence predicts entrepreneurship but not population ruling out scale-based explanations. Second, persistence predicts education, and education predicts firm births consistent with a human capital channel. Third, 1890

church density alone predicts nothing, but long-run continuity does, pointing to accumulation over time rather than a one-time endowment. Together, these patterns are consistent with a model in which persistent civic institutions compounded investments in human capital and social infrastructure across generations, gradually building local entrepreneurial capacity in places that lacked the scale advantages of urban agglomerations.

Schools, skills, and the long arc of learning. The earliest and most direct channel is educational. Congregations historically financed local schools, ran Sunday schools and literacy programs, and normalized study as a communal practice. These activities were small in any one year but relentless across decades. By repeatedly investing in youth and valorizing credential attainment, persistent congregations nudged counties along higher educational trajectories. This is consistent with the evidence in Section 4: persistence predicts modern educational attainment, and both historical schooling and contemporaneous education are strongly associated with new establishment births. The attenuation of the persistence coefficient when education enters the new establishment regression is exactly what one expects if part of the persistence effect is carried by human capital.¹

Trust, norms, and information networks. Churches also produced economic benefits that are not captured by years of schooling alone. Regular, repeated interaction within congregations builds bonding social capital-dense ties that sustain mutual aid, enforce informal contracts, and reduce coordination costs for small ventures. Leaders curate information networks: job leads, credit introductions, cofounder matches, and reputational checks. These mechanisms lower search frictions and facilitate entry at the margin, particularly where formal market infrastructure is thin. In this sense, congregations function as low cost “clearinghouses” for trust and information, translating existing talent into viable enterprise formation.

Substitutes for missing civic infrastructure. Where public capacity was limited, religious organizations substituted for, or scaffolded civic functions: meeting space, youth programming, adult

¹The mediation is partial, not total, which is also informative: churches appear to shape both skills and other non-cognitive or relational inputs to entrepreneurship.

education, and social insurance. This substitution explains why the persistence effect is most visible on the intensive margin of entrepreneurship rather than on the extensive margins of population and income. Counties did not necessarily attract large inflows of people or high wage employers; instead, they learned to use what they had more effectively. The negative association between persistence and population is therefore not paradoxical: stable institutions may raise the returns to staying and building locally, even as urban agglomerations elsewhere absorb population growth.

Reconciling the Petach and Powell Finding The results here may appear to conflict with Petach and Powell (2023), who find that higher religiosity in U.S. counties from 2000 to 2020 is associated with lower GDP growth. The conflict is more apparent than real. Petach and Powell measure current-period church density and adherence — a contemporaneous level of religious activity. The persistence index measures whether religious institutions were continuously present across 130 years. These are distinct objects capturing distinct phenomena. Current church density may reduce GDP growth by stabilizing incumbent firms and slowing the entry-exit churning that drives productivity gains. Institutional persistence may build, over much longer horizons, the accumulated human capital and civic capacity that make new firm formation possible in the first place. Both can be true simultaneously: a high-persistence county with strong civic accumulation may also, in the short run, exhibit the stabilizing effects Petach and Powell document. The present paper does not challenge their finding; it adds a temporal dimension that their cross-sectional religiosity measure cannot capture. Taken together, the two sets of results suggest that the economic effects of religious institutions depend critically on the time horizon over which they are assessed.

Situating the evidence in prior work. The mechanism echoes a classic insight: religious institutions can matter economically through education and norms. Becker and Woessmann argue that the Protestant advantage in 19th-century Germany operated primarily via literacy and schooling rather than theology per se (Becker and Woessmann, 2009). Putnam documents how dense connected life (religious and secular) accumulates civic capacity with durable economic spillovers (Putnam, 1993). Closer to the U.S. county context, recent work emphasizes a trade-off: Petach

and Powell find that religious social capital bolsters firm survival yet may dampen productivity growth by muting entry and exit (Petach and Powell, 2023). The present results complement that picture. Instead of focusing on short run religiosity, the persistence lens highlights a slow moving, place-based asset that raises entrepreneurial intensity per-capita, especially in rural counties where churches were core civic anchors. The attenuation via education links the persistence effect to the formation of skills; the remaining direct effect plausibly reflects trust, norms, and networks that schooling does not fully capture.

Falsifiable implications and scope. The mechanism yields clear testable predictions:

- (i) The persistence effect should be stronger in rural counties and where secular civic infrastructure was historically sparse.
- (ii) It should be larger where denominations historically emphasized schooling and literacy.
- (iii) It should be more closely tied to sectors that rely on local information and trust (e.g., construction, services, small retail).
- (iv) It should be more pronounced when measured against per-capita outcomes than against total counts.

The partial mediation implies that augmenting models with measures of civic engagement (volunteering rates, association density) should further attenuate, but not eliminate, the persistence coefficient. These are not identification strategies per se, but they are ways to probe the channel.

Interpretation and limits The design is descriptive: persistence is a slow-moving institutional summary, not a randomly assigned treatment. Nonetheless, the pattern that education and new establishment births moves around persistence, and population moves inversely fits a coherent, historically grounded mechanism in which churches acted as patient, embedded producers of human capital and trust. In that sense, the paper’s contribution is not that churches “caused” en-

trepreneurship, but that where religious institutions endured, they left a measurable imprint on how communities organized economic life: fewer people, perhaps, but more enterprise per person.

7 Counterarguments and Limitations

Any empirical link between institutions and economic outcomes must be interpreted cautiously. Several counterarguments deserve consideration, both to clarify what the results do and do not show, and to situate the findings within their historical limits.

Reverse causality One objection is that causality may run in the opposite direction: perhaps counties with slower economic growth retained more churches, not because churches promoted entrepreneurship, but because weaker labor markets discouraged out migration and preserved religious infrastructure. In this view, persistence is simply a byproduct of economic stagnation. While this is plausible, the century long horizon of the analysis (1890–2023) tempers the concern. The persistence index captures institutional continuity across multiple generations, well beyond short-run growth cycles. Even so, it remains possible that part of the measured effect reflects demographic inertia rather than institutional influence.

Unobserved cultural confounders A second critique is that persistence proxies for deeper cultural traits such as political conservatism, migration selectivity, or other forms of social capital that are themselves correlated with entrepreneurship. State fixed effects absorb broad regional differences, but they cannot account for all within state heterogeneity. Counties with high persistence may also be places where informal norms or political traditions shaped entrepreneurship independently of religious institutions. While this possibility cannot be fully dismissed, the robustness exercises (Section 6) suggest that the persistence effect is not reducible to urbanization, or population scale.

Religious heterogeneity A further limitation is that the persistence index aggregates across denominations, treating “religious institutions” as equivalent. In practice, traditions differed sharply in their educational, social, and economic functions. The case of Utah illustrates this point: Mormon persistence coincides with strong human capital investment and entrepreneurial outcomes, a combination not necessarily replicable elsewhere. By contrast, Catholic continuity in rural Italy or even Vatican City itself shows that dense and durable religious institutions can persist without generating economic dynamism.

Index limits A related concern is that the persistence index may still reflect uneven data quality across decades, particularly for the 1890 church census, which was compiled under less standardized definitions and coverage than later surveys. Such discrepancies could introduce measurement error and inflate apparent persistence. However, the index construction explicitly guards against this: each decade’s church density is median–standardized, trimmed, and weighted by its cross-decade agreement, reducing the influence of outlier years. Moreover, results remain positive and significant when the index is restricted to modern benchmarks (1990–2020) or when only the early two decades (1890–1990) are used. This consistency suggests that the persistence effect is not an artifact of any single data year or normalization choice, but reflects genuine continuity of institutional presence across time.

Causal limits Finally, the results should not be interpreted as strict causal effects. Even with extensive historical controls and fixed effects, the analysis remains observational. A truly causal design would require quasi-experimental variation in church persistence that is plausibly exogenous to economic outcomes. Several identification strategies are potentially feasible in future work. Rapid migration waves, particularly those driven by push factors in origin counties differentially disrupted local church networks and could serve as instruments for persistence. Large church consolidations and denomination mergers in the mid-twentieth century similarly altered local religious infrastructure for reasons partly independent of economic conditions. Natural disasters that destroyed congregational stock, and the staggered adoption of blue laws that altered the social

functions of churches, offer additional variation. The present estimates should be read as precisely estimated descriptive associations, consistent with a well-specified historical mechanism, and as a foundation for the identification designs that future data collection will make possible.

Interpretation Taken together, these limitations suggest caution but not dismissal. The evidence indicates that where religious institutions endured, they left a measurable imprint on modern economic life, particularly on the intensive margin of entrepreneurship. Yet persistence is not destiny. Institutional continuity can foster skills, trust, and networks, but whether these translate into durable growth depends on context: not all traditions or localities harnessed persistence in the same way. The results should therefore be seen as suggestive evidence of a long-run institutional channel, not as a claim that churches universally or mechanically produced entrepreneurship.

8 Conclusion

This paper has traced a simple but durable pattern in U.S. economic life: places where religious institutions endured across generations built more enterprise per person. Counties with higher church persistence create more new establishment births per-capita today, even though they are not larger and are not systematically richer. The specification remains positive when conditioning on contemporary church density, and is not explained by initial (1890) density alone. Together with the negative association between persistence and population, the evidence points to growth that is upward, not outward: institutional continuity appears to raise the intensity of local economic activity rather than the scale.

The mechanism evidence is consistent with a human capital channel. Persistence strongly predicts modern educational attainment, historical schooling is tightly linked to new establishment births, and the persistence coefficient attenuates when contemporary education is included in the new establishments regression (with a modest, nonzero share mediated). That pattern fits a narrative in which long lived congregations accumulated skills and social capacity week after week, cohort after cohort, lowering the frictions of entry for small enterprises. Trust, norms of reliability,

and information networks likely contribute to the remaining direct association that education alone does not capture.

An important implication of the results is that initial church density in 1890 does not predict modern entrepreneurship once long-run continuity is taken into account. This stands in contrast to much of the deep-roots literature, which emphasizes initial conditions as the key determinant of development (e.g. Becker and Woessmann 2009; Nunn 2009). The evidence here suggests instead that it is institutional persistence across generations, not one-time intensity, that shapes local entrepreneurial capacity.

Methodologically, the multi-period persistence index averaging standardized church density from 1890 to 2020, weighted by cross-decade agreement operationalizes institutional continuity in a way that separates it from both initial conditions and contemporary levels. This measurement strategy is applicable beyond religious institutions: any repeated, spatially granular measure of civic organizational presence could be indexed in this way. Substantively, the focus on new establishment births highlights an intensive margin of local development that is easy to miss in GDP aggregates but central to how many rural counties actually sustain economic life across generations.

The results suggest that community building organizations, those that meet regularly, transmit expectations across generations, and provide stable platforms for learning and coordination can be developmental assets, particularly where scale is limited. While the evidence here concerns churches, the logic may extend to other place bound institutions: libraries, extension offices, community colleges, civic clubs, youth programs, and apprenticeship networks. Interventions that cultivate human capital, reduce search and trust frictions, and preserve “meeting infrastructure” are likely complements, not substitutes, for classic economic development tools.

At the same time, scope and limits matter. The analysis is observational; despite extensive historical controls and fixed effects, the estimates are descriptive, not point identified causal effects. Results should therefore be considered as suggestive. Institutional heterogeneity also matters: different traditions and local histories likely produce different economic footprints. Data constraints

(e.g., incomplete early coverage in a few states) further counsel humility.

While this paper examines religious institutions specifically, the mechanisms identified, human capital accumulation, trust-building, and civic coordination may generalize to other persistent local organizations that perform similar community functions.

Future work can push on both measurement and identification. On measurement, extending the persistence concept to other institutions schools, civic associations, cooperative lenders would test whether the pattern is general. On identification, shocks that differentially altered local religious infrastructure (e.g., rapid migration waves, large church consolidations, blue-law changes, or natural disasters that destroyed/rebuilt congregational stock) could support quasi-experimental designs. Linking county outcomes to micro data on firm founding teams, credit relationships, or volunteer networks would also illuminate the channels more directly.

Economic development is most often told as a story of scale, more people, more capital, bigger markets. The evidence here complicates that story. Communities that endured, that maintained their civic infrastructure across generations without the benefit of population growth or urban agglomeration, were not development failures awaiting rescue. They were accumulating something that scale cannot easily replicate: the trust, the skills, and the institutional memory that make entrepreneurial activity possible even in thin markets. That, too, is development — and measuring it requires looking not at how large a place grew, but at how deeply it learned to use what it had.

Disclosure Statement

The author declares no relevant financial or personal conflicts of interest.

Ethics/IRB

This study uses publicly available aggregate data.

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A Appendix A: Regression Tables

Table A1 Drop-One-Control Exercises (Variants of baseline)

Model	Persistence coef (SE)	R^2
Baseline (A1)	0.0559*** (0.014)	0.954
No Income 1989 (std)	0.0475*** (0.014)	0.953
No Education 1990 (std)	0.0485*** (0.014)	0.950
No log(Pop) 2010	-0.8241*** (0.028)	0.695
State fixed effects	Yes	
Observations	2,968	

Notes: HC3 robust SEs in parentheses. “(std)” variables standardized to mean 0, SD 1. Each row removes only the listed control(s); all other baseline controls and state fixed effects are retained. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A2 Bundled Exclusion: Pairs of Controls Removed (Variants of baseline)

Model	Persistence coef (SE)	R^2	N
No Income 1989 (std) & No Education 1990 (std)	0.0081 (0.015)	0.945	2,968
State fixed effects	Yes		

Notes: HC3 robust standard errors in parentheses. Variables labeled “(std)” are standardized to mean 0 and SD 1 prior to estimation. Each row removes the listed pair of controls; all other baseline controls and state fixed effects are retained. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3 Alternative Institutional Measure: Adding Church Density₂₀₂₀ (Variants of baseline)

OLS: $\log(1 + \text{NewEstab}_{2022})$	
Persistence Index	0.1002*** (0.028)
Church Density 2020 (per 1,000)	-0.0334 (0.019)
Income 1989 (std)	0.0673*** (0.012)
Educational Attainment 1990 (std)	0.2100*** (0.018)
$\log(\text{Population } 2010)$	1.0196*** (0.010)
State fixed effects	Yes
Observations	2,968
R^2	0.954

Notes: HC3 robust standard errors in parentheses. Variables labeled “(std)” are standardized to mean 0 and SD 1 prior to estimation. All models include state fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4 Pop.₁₈₉₀ in place of pop.₂₀₁₀ (Variant of baseline)

OLS: $\log(1 + \text{NewEstab}_{2022})$	
Persistence Index	-0.6986*** (0.029)
Income 1989 (std)	0.4463*** (0.032)
Educational Attainment 1990 (std)	0.2343*** (0.044)
$\log(\text{Population } 1890)$	0.3917*** (0.030)
State fixed effects	Yes
Observations	2,652
R^2	0.756

Notes: HC3 robust standard errors in parentheses. Variables labeled “(std)” are standardized to mean 0 and SD 1 prior to estimation. All models include state fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5 No State Fixed Effects (Variants of baseline)

	OLS: $\log(1 + \text{NewEstab_2022})$
Persistence Index	0.0205 (0.013)
Income 1989 (std)	0.0725*** (0.011)
Educational Attainment 1990 (std)	0.1776*** (0.011)
$\log(\text{Population 2010})$	0.9893*** (0.009)
State fixed effects	No
Observations	2,968
R^2	0.936

Notes: HC3 robust standard errors in parentheses. Variables labeled “(std)” are standardized to mean 0 and SD 1 prior to estimation. Specification excludes state fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6 Alternative Index Construction (1890-1990)

	OLS: $\log(1 + \text{NewEstabs_2022})$
Two Year Persistence Index (1890-1990)	0.0375 (0.011)
Income 1989 (std)	0.0642*** (0.012)
Educational Attainment 1990 (std)	0.2092*** (0.018)
$\log(\text{Population 2010})$	1.0103*** (0.009)
State fixed effects	yes
Observations	2,968
R^2	0.954

Notes: HC3 robust standard errors in parentheses. Variables labeled “(std)” are standardized to mean 0 and SD 1 prior to estimation. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

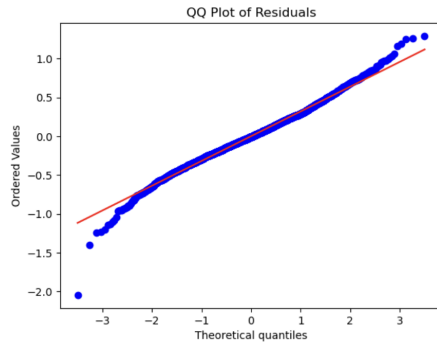
Table A7 The intensive margin: population controls and entrepreneurship (variant of baseline model)

	(1)	(2)
	Contemporary Control	Historical Control
	$\log(\text{Pop}_{2010})$	$\log(\text{Pop}_{1890})$
Persistence Index	0.0559*** (0.014)	-0.6986*** (0.029)
Historical Controls	Yes	Yes
State Fixed Effects	Yes	Yes
Observations	2,968	2,652
R^2	0.954	0.756

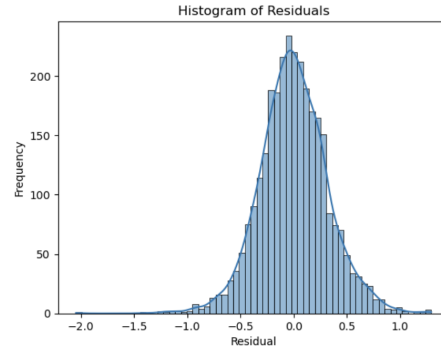
Notes: Both specifications include standardized income (1989), education (1990), and state fixed effects. Robust standard errors in parentheses. *** $p < 0.01$.

B Appendix B: Tables

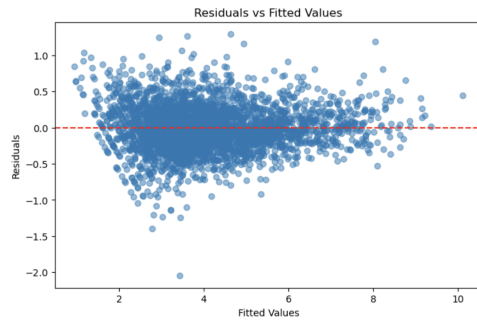
B1 Residual Diagnostics



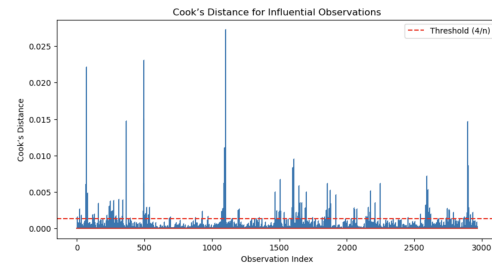
(a) QQ Plot of Residuals



(b) Histogram of Residuals



(c) Residuals vs Fitted



(d) Cook's Distance

Residual diagnostics for the baseline OLS regression of establishment births on persistence and controls. Panels show (a) QQ plot of residuals, (b) histogram of residuals, (c) residuals versus fitted values, and (d) Cook's distance for influential observations. The diagnostics indicate approximate normality, no extreme outliers (all well below conventional 0.5 cutoff), and heteroskedasticity accounted for by robust (HC3) standard errors.

Breusch–Pagan Test Results. LM Statistic = 261.58 ($p < 0.001$), F Statistic = 5.526 ($p < 0.001$).

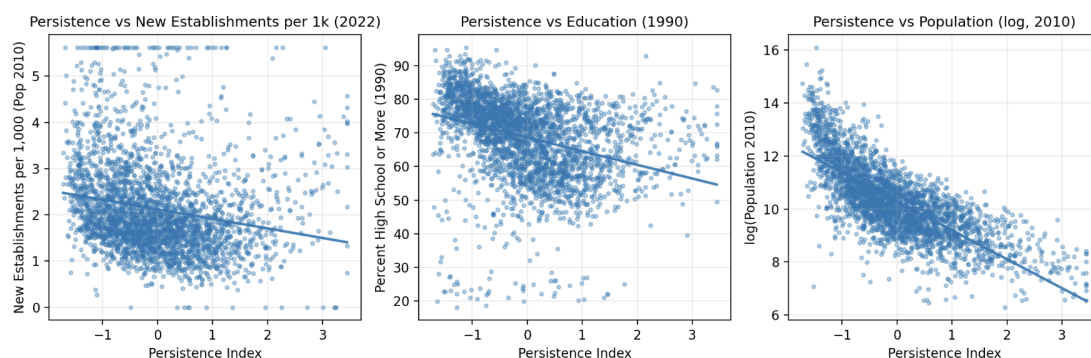
The null hypothesis of homoskedasticity is rejected, indicating heteroskedasticity. All reported

regressions therefore use HC3 heteroskedasticity-robust standard errors.

B2 Summary Statistics

Variable	Mean	Std. Dev.	Min	Max	N
Persistence Index	0.00	1.00	-1.72	3.44	3,055
Persistence Index (1890–1990)	0.00	1.00	-2.23	2.48	3,055
New Establishments, 2022	273.5	1,113.2	0.00	38,467.0	3,004
Population, 1890	22,257.55	54,068.18	0	1,515,301	2,716
Population, 2010	98,232.7	315,733.8	90.0	9,818,605.0	3,055
Population, 2023	106,664.0	335,829.0	81.0	9,663,345.0	3,055
Median Household Income, 1989 (real, \$2023)	58,136.0	15,572.0	21,016.0	144,959.0	3,055
Educational Attainment, 1990 (%)	68.80	11.90	18.10	95.50	3,055
New Establishments, 1989	231.4	837.9	0.00	26,540.0	3,013

B3 Scatter plot diagnostics



Raw bivariate scatterplots of the persistence index against three outcomes: new establishment births per 1,000 residents in 2022 (scaled by Population 2010), percent high school or more in 1990, and log population in 2010. Each line shows the fitted bivariate OLS slope.

The bivariate plots show clear selection patterns. The 3rd indicates a strong negative relationship between the persistence index and $\log(\text{Pop}_{2010})$: places with high persistence tend to be smaller counties. Panels 1 and 2 also slope downward, meaning that, without controls, higher persistence is associated with lower establishment births per capita and lower high school attainment in 1990. These raw patterns reflect composition, especially county size and historical human capital, rather than the partial effect of persistence. In the baseline OLS that conditions on 1989

income, 1990 education, and $\log(\text{Pop}_{2010})$ with state fixed effects, the coefficient on persistence is positive and statistically significant. Thus, after netting out scale and baseline fundamentals, institutional persistence is associated with more new establishments today.

Appendix D: Data Coverage and Missing contemporary Counties

Full List of Counties Missing from Analytic Dataset

State	County	FIPS
AK		
	Aleutians East Borough	02013
	Aleutians West Census Area	02016
	Anchorage Municipality	02020
	Bethel Census Area	02050
	Bristol Bay Borough	02060
	Denali Borough	02068
	Dillingham Census Area	02070
	Fairbanks North Star Borough	02090
	Haines Borough	02100
	Hoonah-Angoon Census Area	02105
	Juneau City and Borough	02110
	Kenai Peninsula Borough	02122
	Ketchikan Gateway Borough	02130
	Kodiak Island Borough	02150
	Kusilvak Census Area	02158

(continued on next page)

State	County	FIPS
	Lake and Peninsula Borough	02164
	Matanuska-Susitna Borough	02170
	Nome Census Area	02180
	North Slope Borough	02185
	Northwest Arctic Borough	02188
	Petersburg Borough	02195
	Prince of Wales-Hyder Census Area	02198
	Sitka City and Borough	02220
	Skagway Municipality	02230
	Southeast Fairbanks Census Area	02240
	Valdez-Cordova Census Area	02261
	Wade Hampton Census Area	02270
	Wrangell City and Borough	02275
	Yakutat City and Borough	02282
	Yukon-Koyukuk Census Area	02290
CO		
	Broomfield County	08014
CT		
	Fairfield County	09001
	Hartford County	09003
	Litchfield County	09005
	Middlesex County	09007
	New Haven County	09009
<i>(continued on next page)</i>		

State	County	FIPS
	New London County	09011
	Tolland County	09013
	Windham County	09015
RI		
	Bristol County	44001
	Kent County	44003
	Newport County	44005
	Providence County	44007
	Washington County	44009
DC		
	District of Columbia	11001
GA		
	Dade County	13083
MO		
	Dade County	29057
NM		
	Doña Ana County	35013
NV		
	Esmeralda County	32009
SD		
	Oglala Lakota County	46102
	Shannon County	46113

(continued on next page)

State	County	FIPS
TX		
	Kenedy County	48261
	Loving County	48301